N19 Wigner transport equation

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Please see appendix D of part II thesis for further detail on Dirac notation in mode space. Mode operators space are denoted by tilde, quantum operators are denoted by hat. The time evolution of a mixed state $\hat{\rho}$ is governed by von-Neumann equation

$$i\hbar\frac{\partial\hat{\rho}}{\partial t} = [\hat{H}, \hat{\rho}] \tag{1.1}$$

The Hamiltonian is split into harmonic $\hat{H}_0$ and anharmonic part $\hat{V}'$

$$\hat{H} = \hat{H}_0 + \hat{V}' \tag{1.2}$$

using notation in thesis appendix, we can write

$$\hat{H}_0 = \hbar\langle\hat{a}|\tilde{\omega}|\hat{a}\rangle + E_0 \tag{1.3}$$

where E_0 is the sum of zero point energy in all phonon modes. Phonon mode with joint index $k = (\mathbf{q}, s)$ which forms a complete set in mode space

$$\tilde{1} = \sum_k |k\rangle\langle k| \tag{1.4}$$

The typical ladder operators are inner products in mode space,

$$\hat{a}_k = \langle k|\hat{a}\rangle, \qquad \hat{a}_k^\dagger = \langle\hat{a}|k\rangle \tag{1.5}$$

which satisfy

$$[\hat{a}_k, \hat{a}_{k'}^\dagger] = \delta_{kk'} \tag{1.6}$$

and $\tilde{\omega}$ has the spectral expansion

$$\tilde{\omega} = \sum_k |k\rangle\omega_k\langle k| \tag{1.7}$$

the commutator

$$[\hat{a}_k, \hat{H}_0] = \hbar\omega_k\hat{a}_k \tag{1.8}$$

de-project out of k-representation gives

$$[|\hat{a}\rangle, \hat{H}_0] = \hbar\tilde{\omega}|\hat{a}\rangle \tag{1.9}$$

similarly

$$[\langle\hat{a}|, \hat{H}_0] = -\hbar\langle\hat{a}|\tilde{\omega} \tag{1.10}$$

where the commutator is generalised for operators in the mode-state product space, then

$$[|\hat{a}\rangle\langle\hat{a}|, \hat{H}_0] = [|\hat{a}\rangle, \hat{H}_0]\langle\hat{a}| + |\hat{a}\rangle[\langle\hat{a}|, \hat{H}_0] = \hbar[\tilde{\omega}, |\hat{a}\rangle\langle\hat{a}|] \tag{1.11}$$

define

$$\tilde{n}(t) \equiv \mathrm{tr}\,[\hat{\rho}(t)|\hat{a}\rangle\langle\hat{a}|] - \tilde{1} \tag{1.12}$$

and

$$\tilde{S}(t) \equiv \frac{i}{\hbar}\,\mathrm{tr}\Big[\hat{\rho}(t)[\hat{V}', |\hat{a}\rangle\langle\hat{a}|]\Big] \tag{1.13}$$

where tr denotes partial trace over state space, then we have

$$\begin{aligned}
\frac{\partial\tilde{n}}{\partial t} &= \mathrm{tr}\left[\frac{\partial\hat{\rho}}{\partial t}|\hat{a}\rangle\langle\hat{a}|\right] \\
&= \frac{1}{i\hbar}\,\mathrm{tr}\left[\hat{\rho}(t)[|\hat{a}\rangle\langle\hat{a}|, \hat{H}]\right] \\
&= -i\,\mathrm{tr}\,[\hat{\rho}(t)[\tilde{\omega}, |\hat{a}\rangle\langle\hat{a}|]] + \frac{i}{\hbar}\,\mathrm{tr}[\hat{\rho}(t)[V', |\hat{a}\rangle\langle\hat{a}|]] \\
&= -i[\tilde{\omega}, \tilde{n}(t) + \tilde{1}] + \tilde{S}(t) \\
&= -i[\tilde{\omega}, \tilde{n}(t)] + \tilde{S}(t)
\end{aligned} \tag{1.14}$$

thus

$$\frac{\partial\tilde{n}}{\partial t} + i[\tilde{\omega}, \tilde{n}(t)] = \tilde{S}(t) \tag{1.15}$$

which is reminiscent of Boltzmann equation, however, no approximation has been made so far and Eq.(15) is exact quantum dynamics. Now note that the $\tilde{\omega}$ can be put into block diagonal form

$$\tilde{\omega} = \bigoplus_{\mathbf{q}} \bar{\omega}_{\mathbf{q}} \tag{1.16}$$

where overhead bar denotes operation into mode subspace for a given $\mathbf{q}$. Define Wigner transform

$$\bar{n}_{\mathbf{q}}(\mathbf{r}, t) = \sum_{\mathbf{k}} \left\langle \mathbf{q} + \frac{\mathbf{k}}{2}|\tilde{n}(t)|\mathbf{q} - \frac{\mathbf{k}}{2}\right\rangle e^{i\mathbf{k}\cdot\mathbf{r}} = \sum_{\mathbf{k}} \bar{f}_{\mathbf{q}}(\mathbf{k}, t)e^{i\mathbf{k}\cdot\mathbf{r}} \tag{1.17}$$

and similarly

$$\bar{S}_{\mathbf{q}}(\mathbf{r}, t) = \sum_{\mathbf{k}} \left\langle \mathbf{q} + \frac{\mathbf{k}}{2}|\tilde{S}(t)|\mathbf{q} - \frac{\mathbf{k}}{2}\right\rangle e^{i\mathbf{k}\cdot\mathbf{r}} = \sum_{\mathbf{k}} \bar{S}_{f,\mathbf{q}}(\mathbf{k}, t)e^{i\mathbf{k}\cdot\mathbf{r}} \tag{1.18}$$

Squeez Eq.(15) between $\langle\mathbf{q} + \frac{\mathbf{k}}{2}|$ and $|\mathbf{q} + \frac{\mathbf{k}}{2}\rangle$ and using the fact that these are eigenmodes of $\tilde{\omega}$ gives

$$\frac{\partial\bar{f}}{\partial t} + i\bar{\omega}_{\mathbf{q}+\mathbf{k}/2}\bar{f} - i\bar{f}\bar{\omega}_{\mathbf{q}-\mathbf{k}/2} = \bar{S}_f \tag{1.19}$$

now the first approximation

$$\bar{\omega}_{\mathbf{q}\pm\mathbf{k}/2} \simeq \bar{\omega}_{\mathbf{q}} \pm \frac{1}{2}\sum_{\alpha} k_\alpha \frac{\partial\bar{\omega}_{\mathbf{q}}}{\partial q_\alpha} \tag{1.20}$$

substitute back gives

$$\frac{\partial\bar{f}}{\partial t} + i[\bar{\omega}, \bar{f}] + \frac{1}{2}\sum_{\alpha} k_\alpha \left\{\frac{\partial\bar{\omega}}{\partial q_\alpha}, \bar{f}\right\} = \bar{S}_f \tag{1.21}$$

where curly braces is a anti-commutator $\{A, B\} = AB + BA$. Multiply both sides by $e^{i\mathbf{k}\cdot\mathbf{r}}$ then sum over $\mathbf{k}$ gives

$$\frac{\partial \bar{n}}{\partial t} + i[\bar{\omega}, \bar{n}] + \frac{1}{2}\sum_{\mathbf{k}}\sum_{\alpha} ik_\alpha e^{i\mathbf{k}\cdot\mathbf{r}} \left\{ \frac{\partial \bar{\omega}}{\partial q_\alpha}, \bar{f} \right\} = \bar{S} \tag{1.22}$$

now use

$$\frac{\partial \bar{n}}{\partial r_\alpha} = \sum_{\mathbf{k}} ik_\alpha e^{i\mathbf{k}\cdot\mathbf{r}} \bar{f} \tag{1.23}$$

and define

$$\bar{v}_{\mathbf{q},\alpha} \equiv \frac{\partial \bar{\omega}_{\mathbf{q}}}{\partial q_\alpha} \tag{1.24}$$

we have

$$\frac{\partial \bar{n}}{\partial t} + i[\bar{\omega}, \bar{n}] + \frac{1}{2}\sum_{\alpha} \left\{ \bar{v}_\alpha, \frac{\partial \bar{n}}{\partial r_\alpha} \right\} = \bar{S} \tag{1.25}$$

in vector notation

$$\boxed{\frac{\partial \bar{n}}{\partial t} + i[\bar{\omega}, \bar{n}] + \frac{1}{2}\{\bar{\mathbf{v}}, \cdot\nabla \bar{n}\} = \bar{S}} \tag{1.26}$$

this is the **Wigner transport equation** (WTE), it is quantum dynamics under the approximation Eq.(20). This is same as Eq.(6) of Simoncelli et al. Nat. Physics WTE paper. Now squeeze the equation between $\langle s|$ and $|s'\rangle$ inserting a complete set of state

$$\bar{1} = \sum_s |s\rangle\langle s| \tag{1.27}$$

between $\bar{\mathbf{v}}$ and $\nabla\bar{n}$ and note that ∇ operates into spacial degree of freedom not mode degree of freedom and recognising

$$S_{\mathbf{q},ss'} = \left(\frac{\partial n_{\mathbf{q},ss'}}{\partial t}\right)_{\mathrm{c}} \tag{1.28}$$

is a collision term, gives Eq.(37) of Simoncelli et al. PRX WTE paper

$$\frac{\partial n_{ss'}}{\partial t} + i\left(\omega_s n_{ss'} - n_{ss'}\omega_{s'}\right) + \frac{1}{2}\sum_{s''}\left(\mathbf{v}_{ss''}\cdot\nabla n_{s''s'} + \nabla n_{ss''}\cdot\mathbf{v}_{s''s'}\right) = \left(\frac{\partial n_{ss'}}{\partial t}\right)_{\mathrm{c}} \tag{1.29}$$

in the case where $n_{ss'} = n_{ss}\delta_{ss'}$, we have for each $\mathbf{q}s$

$$\frac{\partial n}{\partial t} + \mathbf{v}\cdot\nabla n = \left(\frac{\partial n}{\partial t}\right)_{\mathrm{c}} \tag{1.30}$$

which is the **Boltzmann transport equation** (BTE). Where

$$\begin{aligned}
n_{\mathbf{q},ss'}(t) &\equiv \langle s|\bar{n}|s'\rangle \\
&= \sum_{\mathbf{k}} \left\langle \mathbf{q}+\frac{\mathbf{k}}{2}, s|\tilde{n}(t)|\mathbf{q}-\frac{\mathbf{k}}{2}, s' \right\rangle e^{i\mathbf{k}\cdot\mathbf{r}} \\
&= \sum_{\mathbf{k}} \left(\mathrm{tr}\left[\hat{\rho}(t)\hat{a}_{\mathbf{q}+\mathbf{k}/2,s}\hat{a}^\dagger_{\mathbf{q}-\mathbf{k}/2,s'}\right] - \delta_{\mathbf{q}+\mathbf{k}/2,\mathbf{q}-\mathbf{k}/2}\delta_{ss'} \right) e^{i\mathbf{k}\cdot\mathbf{r}} \\
&= \sum_{\mathbf{k}} \mathrm{tr}\left[\hat{\rho}(t)\hat{a}^\dagger_{\mathbf{q}-\mathbf{k}/2,s'}\hat{a}_{\mathbf{q}+\mathbf{k}/2,s}\right] e^{i\mathbf{k}\cdot\mathbf{r}} \\
&= \sum_{\mathbf{k}} \langle \hat{a}^\dagger_{\mathbf{q}-\mathbf{k}/2,s'}\hat{a}_{\mathbf{q}+\mathbf{k}/2,s}\rangle_t e^{i\mathbf{k}\cdot\mathbf{r}}
\end{aligned} \tag{1.31}$$

which is Eq.(5) of Simoncelli et al. Nat. Physics WTE papaer. Immediately, we have $n_{\mathbf{q},ss'} = n^*_{\mathbf{q},s's}$, thus $\bar{n}_{\mathbf{q}} = \bar{n}^\dagger_{\mathbf{q}}$ is Hermitian. When $s = s'$

$$n_{\mathbf{q}s}(t) = \sum_{\mathbf{k}} \langle \hat{a}^\dagger_{\mathbf{q}-\mathbf{k}/2,s} \hat{a}_{\mathbf{q}+\mathbf{k}/2,s} \rangle_t e^{i\mathbf{k}\cdot\mathbf{r}} \tag{1.32}$$

A phonon is an excitation of a normal mode, thus its position in real space is undefined. The Boltzmann equation should therefore be understood semiclassically: it describes phonon wavepackets that are sufficiently localised in real space while remaining sufficiently narrow in reciprocal space. Each phonon wavepacket is constructed from a superposition of phonon eigenstates near $\mathbf{q}$. By similar argument, the scattering term is

$$\left(\frac{\partial n_{\mathbf{q},ss'}}{\partial t}\right)_{\mathrm{c}} = \frac{i}{\hbar} \sum_{\mathbf{k}} \mathrm{tr}\Big[\hat{\rho}(t)[\hat{V}', \hat{a}_{\mathbf{q}+\mathbf{k}/2,s} \hat{a}^\dagger_{\mathbf{q}-\mathbf{k}/2,s'}]\Big] e^{i\mathbf{k}\cdot\mathbf{r}} \tag{1.33}$$

which can also be written as

$$\left(\frac{\partial n_{\mathbf{q},ss'}}{\partial t}\right)_{\mathrm{c}} = \frac{i}{\hbar} \sum_{\mathbf{k}} \mathrm{tr}\Big[\hat{\rho}(t)[\hat{V}', \hat{a}^\dagger_{\mathbf{q}-\mathbf{k}/2,s'} \hat{a}_{\mathbf{q}+\mathbf{k}/2,s}]\Big] e^{i\mathbf{k}\cdot\mathbf{r}}. \tag{1.34}$$

and lastly

$$\mathbf{v}_{\mathbf{q},ss'} = \langle s|\bar{\mathbf{v}}_{\mathbf{q}}|s'\rangle \tag{1.35}$$

where we emphasis that $|s\rangle$ is in generate not an eigenket of $\bar{\mathbf{v}}_{\mathbf{q}}$, the diagonal component is phonon group velocity.

$$\mathbf{v}_{\mathbf{q}s} \equiv \mathbf{v}_{\mathbf{q},ss} = \langle s|\bar{\mathbf{v}}_{\mathbf{q}}|s\rangle = \nabla_{\mathbf{q}}\omega_{\mathbf{q}s} \tag{1.36}$$

it is easy to show the final equality by differentiating $\bar{\omega}_{\mathbf{q}}|s\rangle = \omega_{\mathbf{q}s}|s\rangle$ with respect to $\mathbf{q}$ then left multiply by $\langle s|$.

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We now connect the Wigner transport equation to thermal conductivity. The harmonic energy density is

$$u(\mathbf{r}, t) = \frac{1}{V} \sum_{\mathbf{q}s} \hbar\omega_{\mathbf{q}s} \left(n_{\mathbf{q}s} + \frac{1}{2}\right) = \frac{\hbar}{V} \sum_{\mathbf{q}} \mathrm{tr}_{\mathbf{q}} \left[\bar{\omega}_{\mathbf{q}} \left(\bar{n}_{\mathbf{q}} + \frac{1}{2}\right)\right] \tag{2.1}$$

where $\mathrm{tr}_{\mathbf{q}}$ denotes partial trace over mode subspace for a given $\mathbf{q}$. multiply both sides of WTE Eq.(1.26) by $\hbar\bar{\omega}_{\mathbf{q}}$, take partial trace $\mathrm{tr}_{\mathbf{q}}$ then sum over $\mathbf{q}$ gives

$$\hbar \sum_{\mathbf{q}} \mathrm{tr}_{\mathbf{q}} \left[\bar{\omega}_{\mathbf{q}} \frac{\partial \bar{n}_{\mathbf{q}}}{\partial t}\right] + i\hbar \sum_{\mathbf{q}} \mathrm{tr}_{\mathbf{q}} \left[\bar{\omega}_{\mathbf{q}}[\bar{\omega}_{\mathbf{q}}, \bar{n}_{\mathbf{q}}]\right] + \frac{1}{2}\hbar \sum_{\mathbf{q}} \mathrm{tr}_{\mathbf{q}} \left[\bar{\omega}_{\mathbf{q}}\{\bar{\mathbf{v}}, \nabla\bar{n}_{\mathbf{q}}\}\right] = \hbar \sum_{\mathbf{q}} \mathrm{tr}_{\mathbf{q}}[\bar{\omega}_{\mathbf{q}} \bar{S}_{\mathbf{q}}] \tag{2.2}$$

but $\mathrm{tr}_{\mathbf{q}}[\bar{\omega}_{\mathbf{q}}[\bar{\omega}_{\mathbf{q}}, \bar{n}_{\mathbf{q}}]] = 0$ using cyclicity of the trace and the collision term on the right hand side vanishes due to total energy conservation, divide everything left by V and using definition Eq.(2.1) gives

$$\frac{\partial u}{\partial t} + \frac{\hbar}{2V} \sum_{\mathbf{q}} \mathrm{tr}_{\mathbf{q}}[\bar{\omega}_{\mathbf{q}}\{\bar{\mathbf{v}}, \nabla\bar{n}_{\mathbf{q}}\}] = 0 \tag{2.3}$$

now consider for a given $\mathbf{q}$, and with index α

$$\sum_\alpha \mathrm{tr}\left[\bar{\omega}\left\{\bar{v}_\alpha, \frac{\partial \bar{n}}{\partial r_\alpha}\right\}\right] = \sum_\alpha \frac{\partial}{\partial r_\alpha}\mathrm{tr}[\{\bar{\omega}, \bar{v}_\alpha\}\bar{n}] = \nabla \cdot \mathrm{tr}[\{\bar{\omega}, \bar{\mathbf{v}}\}\bar{n}] \tag{2.4}$$

substitute back into Eq.(2.3) gives

$$\frac{\partial u}{\partial t} + \nabla \cdot \frac{\hbar}{2V}\sum_{\mathbf{q}} \mathrm{tr}_{\mathbf{q}}[\{\bar{\omega}_{\mathbf{q}}, \bar{\mathbf{v}}_{\mathbf{q}}\}\bar{n}] = 0 \tag{2.5}$$

comparison with continuity equation for local energy conservation

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{j} = 0 \tag{2.6}$$

allows us to identify harmonic heat flux density

$$\mathbf{j} = \frac{\hbar}{2V}\sum_{\mathbf{q}} \mathrm{tr}_{\mathbf{q}}[\{\bar{\omega}_{\mathbf{q}}, \bar{\mathbf{v}}_{\mathbf{q}}\}\bar{n}_{\mathbf{q}}] \tag{2.7}$$

now evaluate the trace and insert an identity operator in subspace

$$\begin{aligned}
\mathbf{j} &= \frac{\hbar}{2V}\sum_{\mathbf{qs}}\sum_{\mathbf{s}'}\langle s|\bar{\omega}_{\mathbf{q}}\bar{\mathbf{v}}_{\mathbf{q}}|s'\rangle\langle s'|\bar{n}_{\mathbf{q}}|s\rangle + \langle s|\bar{\mathbf{v}}_{\mathbf{q}}\bar{\omega}_{\mathbf{q}}|s'\rangle\langle s'|\bar{n}_{\mathbf{q}}|s\rangle \\
&= \frac{\hbar}{2V}\sum_{\mathbf{q}ss'} \omega_{\mathbf{q}s}\mathbf{v}_{\mathbf{q},ss'}n_{\mathbf{q},s's} + \omega_{\mathbf{q}s'}\mathbf{v}_{\mathbf{q},ss'}n_{\mathbf{q},s's} \\
&= \frac{\hbar}{2V}\sum_{\mathbf{q}ss'}(\omega_{\mathbf{q}s} + \omega_{\mathbf{q}s'})\mathbf{v}_{\mathbf{q},ss'}n_{\mathbf{q},s's}
\end{aligned} \tag{2.8}$$

in the case of $n_{\mathbf{q},ss'} = n_{\mathbf{q}s}\delta_{ss'}$, we recover the harmonic population flux

$$\mathbf{j}^{\mathrm{p}} = \frac{1}{V}\sum_{\mathbf{q}s}\hbar\omega_{\mathbf{q}s}\mathbf{v}_{\mathbf{q}s}n_{\mathbf{q}s} \tag{2.9}$$

however, in general, there is an off-diagonal in s, coherence contribution to heat flux. Using the fact that $\bar{\mathbf{v}} = \bar{\mathbf{v}}^\dagger$ which follows from $\bar{\omega} = \bar{\omega}^\dagger$, we can rewrite Eq.(2.8) as

$$\begin{aligned}
\mathbf{j} &= \frac{\hbar}{2V}\sum_{\mathbf{q}ss'} \omega_{\mathbf{q}s}\mathbf{v}_{\mathbf{q},ss'}n_{\mathbf{q},s's} + \frac{\hbar}{2V}\sum_{\mathbf{q}s's} \omega_{\mathbf{q}s}\mathbf{v}_{\mathbf{q},s's}n_{\mathbf{q},ss'} \\
&= \frac{\hbar}{2V}\sum_{\mathbf{q}ss'} \omega_{\mathbf{q}s}\mathbf{v}_{\mathbf{q},ss'}n_{\mathbf{q},s's} + \frac{\hbar}{2V}\sum_{\mathbf{q}ss'} \omega_{\mathbf{q}s}\mathbf{v}^*_{\mathbf{q},ss'}n^*_{\mathbf{q},s's} \\
&= \frac{1}{V}\sum_{\mathbf{q}ss'}\hbar\omega_{\mathbf{q}s}\mathrm{Re}(\mathbf{v}_{\mathbf{q},ss'}n_{\mathbf{q},s's})
\end{aligned} \tag{2.10}$$

the off-diagonal part in s is the coherence contribution to heat flux

$$\mathbf{j}^{\mathrm{c}} = \frac{\hbar}{2V}\sum_{\mathbf{q},s\neq s'}(\omega_{\mathbf{q}s} + \omega_{\mathbf{q}s'})\mathbf{v}_{\mathbf{q},ss'}n_{\mathbf{q},s's} \tag{2.11}$$

and the total heat flux is

$$\mathbf{j} = \mathbf{j}^{\mathrm{p}} + \mathbf{j}^{c} \tag{2.12}$$

now linearised by considering a small departure from local equilibrium

$$\bar{n}_{\mathbf{q}}(\mathbf{r}) \simeq \bar{n}^0_{\mathbf{q}}(\beta(\mathbf{r})) + \delta\bar{n}_{\mathbf{q}}(\mathbf{r}), \qquad \bar{n}^0_{\mathbf{q}} = (e^{\beta\hbar\bar{\omega}_{\mathbf{q}}} + \bar{1})^{-1} \tag{2.13}$$

since $\bar{n}^0$ is a function of $\bar{\omega}$, we have $[\bar{n}, \bar{\omega}] = 0$ thus $[\bar{\omega}, \bar{n}] = [\bar{\omega}, \delta\bar{n}]$ and differentiation with respect to β yields

$$-\frac{\partial \bar{n}^0}{\partial \beta} = \hbar\bar{\omega}\bar{n}^0(\bar{n}^0 + \bar{1}) \equiv \bar{\chi} \tag{2.14}$$

thus

$$\frac{\partial \bar{n}}{\partial r_\alpha} = \frac{\partial \bar{n}^0(\beta)}{\partial \beta}\frac{\partial \beta(\mathbf{r})}{\partial r_\alpha} + \frac{\partial \delta\bar{n}(\mathbf{r})}{\partial r_\alpha} \tag{2.15}$$

$$\frac{\partial \bar{n}}{\partial t} = \frac{\partial \bar{n}^0(\beta)}{\partial \beta}\frac{\partial \beta(\mathbf{r})}{\partial t} + \frac{\partial \delta\bar{n}(\mathbf{r})}{\partial t} \tag{2.16}$$

we now consider steady stat

$$\frac{\partial \beta}{\partial t} = 0, \qquad \frac{\partial \delta\bar{n}}{\partial t} = 0 \tag{2.17}$$

and since $\partial\delta\bar{n}/\partial r_\alpha$ is a second order term, it may be ignored in this linear regime, then WTE becomes

$$i[\bar{\omega}, \delta\bar{n}] - \frac{1}{2}\sum_\alpha \{\bar{v}_\alpha, \bar{\chi}\}\frac{\partial \beta}{\partial r_\alpha} = \bar{S} \tag{2.18}$$

linearising the scattering term

$$\bar{S} \simeq -C[\delta\bar{n}] \tag{2.19}$$

gives the linearised Wigner transport equation (LWTE)

$$\frac{1}{2}\sum_\alpha \{\bar{v}_\alpha, \bar{\chi}\}\frac{\partial \beta}{\partial r_\alpha} = -C[\delta\bar{n}] + i[\bar{\omega}, \delta\bar{n}] \tag{2.20}$$

in vector notation

$$\boxed{\frac{1}{2}\{\bar{\mathbf{v}}, \bar{\chi}\} \cdot \nabla\beta = i[\bar{\omega}, \delta\bar{n}] + C[\delta\bar{n}]} \tag{2.21}$$

now squeeze the equation between $\langle s|$ and $|s'\rangle$ gives

$$\frac{\chi_{\mathbf{q}s} + \chi_{\mathbf{q}s'}}{2}\mathbf{v}_{\mathbf{q},ss'} \cdot \nabla\beta = i(\omega_{\mathbf{q}s} - \omega_{\mathbf{q}s'})\delta n_{\mathbf{q},ss'} + C_{\mathbf{q}.ss'}[\delta\bar{n}] \tag{2.22}$$

where $C_{\mathbf{q},ss'}[\delta\bar{n}] = \langle s|C[\delta\bar{n}]|s'\rangle$. Define population matrix $\Omega_{\mathbf{q}s,\mathbf{q}'s'}$ via

$$\sum_{\mathbf{q}'s'} \Omega_{\mathbf{q}\mathbf{s},\mathbf{q}'\mathbf{s}'}\delta n_{\mathbf{q}'\mathbf{s}'} = \langle s|C[\delta\bar{n}]|s\rangle \tag{2.23}$$

where $\Omega_{\mathbf{q}s,\mathbf{q}'s'}$ is a positive semi-definite matrix to ensure local entropy production, so the second law of termodyanmics is not violated. For off diagonal element $s \neq s'$, Simoncelli approximated

$$\eta_{\mathbf{q},ss'}\delta n_{\mathbf{q},ss'} \simeq \langle s|C[\delta\bar{n}]|s'\rangle, \qquad \eta_{\mathbf{q},ss'} \equiv \frac{\gamma_{\mathbf{q}s} + \gamma_{\mathbf{q}s'}}{2} \tag{2.24}$$

Here $\gamma_{\mathbf{q}s} = \Omega_{\mathbf{q}s,\mathbf{q}s}$ is the population relaxation rate in the single-mode relaxation-time approximation. We approximate the coherence damping rate by half the sum of the two population relaxation rates.

$$\gamma_{\mathbf{q}s} \equiv \Omega_{\mathbf{q}s,\mathbf{q}s} \tag{2.25}$$

the diagonal term in s gives linearised Boltzmann transport equation (LBTE)

$$\chi_{\mathbf{q}s}\mathbf{v}_{\mathbf{q}s} \cdot \nabla\beta = \sum_{\mathbf{q}'s'} \Omega_{\mathbf{q}s,\mathbf{q}'s'}\delta n_{\mathbf{q}'s'} \tag{2.26}$$

which can be written in a more familiar form

$$\frac{\partial n^0_{\mathbf{q}s}}{\partial T}\mathbf{v}_{\mathbf{q}s} \cdot \nabla T = -\sum_{\mathbf{q}'s'} \Omega_{\mathbf{q}s,\mathbf{q}'s'}\delta n_{\mathbf{q}s'} \tag{2.27}$$

the off diagonal part of LWTE is

$$\frac{\chi_{\mathbf{q}s} + \chi_{\mathbf{q}s'}}{2}\mathbf{v}_{\mathbf{q},ss'} \cdot \nabla\beta = (\eta_{\mathbf{q},ss'} + i(\omega_{\mathbf{q}s} - \omega_{\mathbf{q}s'}))\delta n_{\mathbf{q},ss'} \tag{2.28}$$

so $\delta n_{\mathbf{q},ss'}, s \neq s'$ immediately has the following solution

$$\delta n_{\mathbf{q},ss'} = \frac{\chi_{\mathbf{q}s} + \chi_{\mathbf{q}s'}}{\gamma_{\mathbf{q}s} + \gamma_{\mathbf{q}s'} + 2i(\omega_{\mathbf{q}s} - \omega_{\mathbf{q}s'})}\mathbf{v}_{\mathbf{q},ss'} \cdot \nabla\beta, \qquad s \neq s' \tag{2.29}$$

which can also be written as

$$\delta n_{\mathbf{q},ss'} = -\frac{\frac{\partial n^0_{\mathbf{q}s}}{\partial T} + \frac{\partial n^0_{\mathbf{q}s'}}{\partial T}}{\gamma_{\mathbf{q}s} + \gamma_{\mathbf{q}s'} + 2i(\omega_{\mathbf{q}s} - \omega_{\mathbf{q}s'})}\mathbf{v}_{\mathbf{q},ss'} \cdot \nabla T, \qquad s \neq s' \tag{2.30}$$

which is Eq.(46) in Simoncelli PRX WTE paper. Substitute into diagonal component of heat flux then comparing with Fourier's law

$$\mathbf{j} = -\boldsymbol{\kappa}\nabla T \tag{2.31}$$

gives the coherence contribution to thermal conductivity

$$\kappa^{\mathrm{c}}_{\mu\nu} = \frac{\hbar}{2V}\sum_{\mathbf{q},s\neq s'} (\omega_{\mathbf{q}s} + \omega_{\mathbf{q}s'})\frac{\frac{\partial n^0_{\mathbf{q}s}}{\partial T} + \frac{\partial n^0_{\mathbf{q}s'}}{\partial T}}{\gamma_{\mathbf{q}s} + \gamma_{\mathbf{q}s'} + 2i(\omega_{\mathbf{q}s'} - \omega_{\mathbf{q}s})}v^{\mu}_{\mathbf{q},ss'}v^{\nu}_{\mathbf{q},s's}, \qquad s \neq s' \tag{2.32}$$

but if we set $s = s'$, then we have

$$\kappa_{\mu\nu} = \frac{1}{V}\sum_{\mathbf{q}s} c_{\mathbf{q}s}v^{\mu}_{\mathbf{q}s}v^{\nu}_{\mathbf{q}s}\frac{1}{\gamma_{\mathbf{q}s}} \tag{2.33}$$

where $c_{\mathbf{q}s} = \hbar\omega_{\mathbf{q}s}\partial n^0_{\mathbf{q}s}/\partial T$ is the harmonic heat capacity. Eq.(2.33) is relaxation time approximation (RTA) thermal conductivity to Boltzmann transport equation, with relaxation time $1/\gamma_{\mathbf{q}s}$. So Eq.(2.24) is a coherence analogue of the population RTA. However, the diagonal component of WTE is the full BTE without RTA, therefore Eq.(2.27) would not give Eq.(2.33) without the diagonal approximation $\Omega_{\mathbf{q}s,\mathbf{q}'s'} \simeq \gamma_{\mathbf{q}s}\delta_{\mathbf{q}s,\mathbf{q}'s'}$. Our Eq.(2.32) is consistent with Simoncelli et al. Nat. Physics WTE paper Eq.(12), which is

$$\kappa^{\mathrm{c}}_{\mu\nu} = \frac{\hbar^2}{Vk_BT^2}\sum_{\mathbf{q},s\neq s'} \frac{\omega_{\mathbf{q}s} + \omega_{\mathbf{q}s'}}{2}v^{\mu}_{\mathbf{q},ss'}v^{\nu}_{\mathbf{q},s's}\frac{\omega_{\mathbf{q}s}n^0_{\mathbf{q}s}(n^0_{\mathbf{q}s'} + 1) + \omega_{\mathbf{q}s'}n^0_{\mathbf{q}s'}(n^0_{\mathbf{q}s'} + 1)}{4(\omega_{\mathbf{q}s} - \omega_{\mathbf{q}s'})^2 + (\gamma_{\mathbf{q}s} + \gamma_{\mathbf{q}s'})^2}(\gamma_{\mathbf{q}s} + \gamma_{\mathbf{q}s'}) \tag{2.34}$$

which seems complicated, but Eq.(2.24) is perhaps the simplest approximation one can make regarding these coherence contributions to thermal conductivity.